

Blog System Testing Guide

Overview

This guide will help you test the updated blog create/edit functionality with TipTap rich text editor integration.

What Was Fixed

1. BlogDetailPage Rendering (CRITICAL FIX)

File: frontend/src/pages/static/BlogDetailPage.tsx (line 399)

Before:

```
dangerouslySetInnerHTML={{
  __html: blog.content.replace(/\n/g, '<br>').replace(/\*(.*?)\*/g, '<strong>$1</strong>')
}}
```

After:

```
dangerouslySetInnerHTML={{ __html: blog.content }}
```

Why: The regex processing was corrupting TipTap's HTML structure. Images, videos, headings, and formatting were broken. Now TipTap HTML renders properly.

2. Blog Content Display Styles

File: frontend/src/styles/index.css (lines 719-802)

Added: Comprehensive `.blog-content` styles for:

- Headings (h1, h2, h3) with proper sizing and spacing
- Paragraphs with line spacing
- Lists (ul, ol) with indentation
- Blockquotes with left border and background
- Code blocks with dark theme
- Links with hover effects
- Images with shadow and centering
- Videos with responsive sizing
- iFrames (YouTube) with 16:9 aspect ratio
- Text formatting (bold, italic, underline, strikethrough)

3. BlogForm Preview

File: frontend/src/components/admin/BlogForm.tsx (line 335)

Status: Already correctly using `blog-content` class - no changes needed!

Testing Steps

Step 1: Start Development Servers

Backend:

```
cd backend
npm run dev
```

Expected: Server starts on port 5000 (or configured port)

Frontend:

```
cd frontend
npm run dev
```

Expected: Server starts on `http://localhost:3000`

Step 2: Access Admin Blog Page

1. Open browser: `http://localhost:3000/admin/blogs`
2. Login with admin credentials
3. You should see the blog management dashboard

Expected View:

- Stats cards (Total Posts, Published, Drafts, Categories)
- Search and filter options
- Table with existing blogs
- "Create Blog" button

Step 3: Test Blog Creation

3.1 Click "Create Blog" Button

- Modal should open with BlogForm
- TipTap editor should be visible with toolbar

3.2 Fill Basic Information

```
Title: Test Blog - TipTap Rich Content
Slug: (auto-generated or manual)
Excerpt: Testing the new TipTap editor with images, videos, and rich formatting.
Category: Select any
Status: Draft (for testing)
```

3.3 Test Text Formatting

Type the following content:

```
This is a test blog to verify TipTap integration.

Introduction
This is a paragraph with bold text and italic text and underlined text.

Features
Testing various features:
• Bullet point 1
• Bullet point 2
• Bullet point 3

Numbered List:
1. First item
2. Second item
3. Third item

Quote
"This is a blockquote to test styling."

This is inline code: console.log('Hello')

This text is aligned to the center.
This text is aligned to the right.
```

Actions:

1. Type "Introduction" → Select it → Click H2
2. Type "bold text" → Select it → Click Bold (B)
3. Type "italic text" → Select it → Click Italic (I)
4. Type "underlined text" → Select it → Click Underline (U)
5. Click Bullet List icon → Add 3 items
6. Click Numbered List icon → Add 3 items
7. Click Blockquote → Type quote
8. Type `console.log('Hello')` → Select → Click Code (</>)
9. Select center text → Click Center Align
10. Select right text → Click Right Align

Expected Result:

- All formatting should apply instantly
- Toolbar buttons should highlight when active
- Text should look formatted in editor

3.4 Test Image Upload**Steps:**

1. Position cursor where you want image
2. Click **Upload Image** icon (first upload button)
3. Select an image file from computer (JPG, PNG, WebP)
4. Wait for upload

Expected Result:

- Upload progress toast appears
- "Image uploaded successfully" toast
- Image appears in editor
- Image is responsive and centered
- Image has rounded corners

Verify in Cloudinary:

- Go to Cloudinary dashboard
- Check `gema/blogs/content/` folder
- Image should be uploaded there

3.5 Test Image URL**Steps:**

1. Position cursor
2. Click **Image URL** icon (image icon)
3. Enter: `https://picsum.photos/800/400`
4. Click OK

Expected Result:

- Image appears immediately
- Same styling as uploaded image

3.6 Test Video Upload**Steps:**

1. Position cursor
2. Click **Upload Video** icon (red upload icon)
3. Select video file (MP4, WebM, MOV)
4. Wait for upload (may take longer)

Expected Result:

- Upload progress
- "Video uploaded successfully" toast
- HTML5 video player appears
- Player has controls (play/pause, volume, fullscreen)
- Video is responsive

Verify in Cloudinary:

- Check `gema/blogs/content/` folder
- Video should be there

3.7 Test YouTube Embed**Steps:**

1. Position cursor
2. Click **YouTube** icon
3. Paste URL: `https://www.youtube.com/watch?v=dQw4w9WgXcQ`

4. Click OK

Expected Result:

- YouTube iframe appears
- 16:9 aspect ratio
- Responsive sizing
- Can play video in editor preview

3.8 Test Link Creation

Steps:

1. Type: "Visit our website for more"
2. Select "our website"
3. Click **Link** icon (chain)
4. Enter URL: `https://kidrove.com`
5. Click OK

Expected Result:

- Text turns blue and underlined
- Link icon highlights when cursor on link
- Clicking link in preview doesn't navigate (editor mode)

3.9 Complete Blog Creation

Fill remaining fields:

- **Author Name:** Test Author
- **Author Email:** test@example.com
- **Featured Image:** Upload any image
- **Tags:** Add tags: test, tiptap, rich-content
- **Status:** Change to "Published"

Click "Create Blog"

Expected Result:

- Loading spinner
 - Success toast: "Blog created successfully"
 - Modal closes
 - New blog appears in blog list
-

Step 4: Test Blog Editing

4.1 Open Existing Blog

1. In blog list, find the blog you just created
2. Click "Edit" button

Expected Result:

- BlogForm modal opens
- All fields populated with saved data
- TipTap editor shows content with formatting
- Images and videos are visible
- All formatting preserved

4.2 Make Changes

1. Add new heading: "Updated Section"
2. Upload another image
3. Change some text formatting
4. Click "Update Blog"

Expected Result:

- Success toast
- Changes saved
- Modal closes

4.3 Edit Again to Verify

- Open same blog for editing
 - Verify new changes are there
 - All formatting still correct
-

Step 5: Test Blog Display on Frontend

5.1 View Published Blog

1. Note the blog slug (e.g., "test-blog-tiptap-rich-content")
2. Navigate to: `http://localhost:3000/blog/test-blog-tiptap-rich-content`

Expected Result:

- Blog loads without errors
- Featured image displays
- Author info shows
- Tags display

5.2 Verify Content Rendering

Check the following elements:



Headings:

- H1/H2/H3 should have larger font, bold
- Proper spacing above and below
- Dark gray color



Text Formatting:

- **Bold text** appears bold and darker
- *Italic text* appears italic
- Underlined text has underline
- ~~Strikethrough~~ has line through



Lists:

- Bullet points have disc markers
- Numbered lists have numbers
- Proper indentation (ml-6)
- Spacing between items



Blockquotes:

- Left border (blue/primary color)
- Gray background
- Italic text
- Padding left and vertical



Code:

- Inline code has gray background, red text
- Code blocks have dark background, light text
- Proper spacing around blocks



Links:

- Blue color
- Underlined
- Hover changes to darker blue
- Clickable (navigates to URL)



Images:

- Full width responsive
- Rounded corners
- Shadow effect
- Centered
- Proper spacing (my-4)

☒ **Videos:**

- Responsive width
- Rounded corners
- Shadow effect
- Controls visible
- Plays when clicked

☒ **YouTube Embeds:**

- 16:9 aspect ratio
- Responsive width
- Rounded corners
- Shadow effect
- Plays when clicked

☒ **Alignment:**

- Left-aligned text on left
- Center-aligned text centered
- Right-aligned text on right

5.3 Test Responsive Design

Resize browser window:

- Desktop (1920px+): Full width layout
- Tablet (768px-1024px): Adjusted columns
- Mobile (< 768px): Single column, images scale

Expected:

- All content remains readable
- Images scale proportionally
- Videos maintain aspect ratio
- YouTube embeds stay responsive
- No horizontal scrolling

5.4 Test on Different Browsers

Test on:

- Chrome
- Firefox
- Safari (if available)
- Edge

Expected:

- Consistent rendering across browsers
- All media loads
- Styles apply correctly

Step 6: Test Preview Mode

6.1 In BlogForm

1. Create or edit a blog
2. Add content with various formatting
3. Click "**Preview**" tab (if available)

Expected Result:

- Preview shows content exactly as it will appear on frontend
- Uses same `.blog-content` styles
- Images, videos, formatting all visible
- Can switch back to Edit mode

Step 7: Test Edge Cases

7.1 Large Images

- Upload image > 2MB
- Should upload successfully (or show size error)
- Image should be optimized by Cloudinary

7.2 Multiple Images

- Add 5-10 images in one blog
- All should upload and display
- Page should load without lag

7.3 Long Content

- Create blog with 20+ paragraphs
- Mix of headings, lists, images
- Should render smoothly
- No performance issues

7.4 Special Characters

- Add emojis: 🌿 🍷 ✨
- Add accented characters: café, naïve
- Should display correctly

7.5 Empty Content

- Try to save blog with empty content
- Should show validation error: "Content is required"

7.6 Mixed Formatting

- Bold + Italic + Underline same text
- Should apply all three formats

7.7 Nested Lists

- Create list with sub-items
- Should indent properly

Troubleshooting

Issue: Images Not Displaying on Frontend

Check:

1. Browser console for errors
2. Network tab - are images loading?
3. Cloudinary dashboard - are images uploaded?
4. Image URLs in blog content (inspect element)

Solutions:

- Verify Cloudinary credentials in backend `.env`
- Check CORS settings
- Ensure `UPLOAD_PROVIDER=cloudinary`

Issue: Styles Not Applying

Check:

1. Browser DevTools → Elements → Inspect blog content
2. Verify `.blog-content` class is present
3. Check if CSS loaded (Network tab)

Solutions:

- Clear browser cache
- Rebuild frontend: `npm run build`
- Check Tailwind config

Issue: Video Not Playing

Check:

1. Video format (MP4, WebM, MOV supported)
2. File size (check backend limits)
3. Browser codec support

Solutions:

- Convert video to MP4 H.264
- Reduce file size
- Use YouTube embed instead

Issue: Editor Not Loading**Check:**

1. Console errors
2. TipTap packages installed: `npm ls @tiptap/react`
3. JavaScript errors

Solutions:

- Reinstall packages: `npm install`
- Clear `node_modules` and reinstall
- Check for conflicting packages

Issue: Formatting Lost on Save**Check:**

1. Network tab - what data is sent?
2. Database - inspect blog content field
3. Blog content should be HTML string

Solutions:

- Verify BlogForm is using TipTapEditor (not textarea)
- Check `editor.getHTML()` is called
- Inspect saved HTML in database

Success Criteria

Your blog system is working correctly if:

- ☒ Can create blog with rich formatting
- ☒ Can upload images within content
- ☒ Can upload videos within content
- ☒ Can embed YouTube videos
- ☒ All formatting displays correctly on frontend
- ☒ Images are optimized and load fast
- ☒ Videos play correctly
- ☒ Responsive design works on mobile
- ☒ Editing existing blogs preserves formatting
- ☒ Preview matches published view
- ☒ No console errors
- ☒ Styles apply consistently

Testing Checklist

Copy this checklist and check off as you test:

Blog Creation

- ☐ Access admin blog page
- ☐ Click "Create Blog"
- ☐ Add heading (H1, H2, H3)
- ☐ Add bold text
- ☐ Add italic text
- ☐ Add underlined text
- ☐ Create bullet list

- ☐ Create numbered list
- ☐ Add blockquote
- ☐ Add inline code
- ☐ Upload image
- ☐ Add image via URL
- ☐ Upload video
- ☐ Embed YouTube video
- ☐ Add link
- ☐ Change text alignment
- ☐ Fill all required fields
- ☐ Save blog

Blog Editing

- ☐ Open existing blog for edit
- ☐ Verify content loads correctly
- ☐ Make changes
- ☐ Save changes
- ☐ Verify changes persisted

Frontend Display

- ☐ Navigate to blog detail page
- ☐ Headings display correctly
- ☐ Text formatting preserved
- ☐ Lists render properly
- ☐ Blockquotes styled
- ☐ Code blocks styled
- ☐ Links are clickable
- ☐ Images display and are responsive
- ☐ Videos play
- ☐ YouTube embeds work
- ☐ Mobile responsive
- ☐ No console errors

Preview Mode

- ☐ Preview shows formatted content
- ☐ Preview matches frontend

Edge Cases

- ☐ Large images upload
- ☐ Multiple media files work
- ☐ Long content renders
- ☐ Special characters display
- ☐ Validation prevents empty content
- ☐ Mixed formatting works
- ☐ Nested lists work

Performance Benchmarks

Expected Load Times:

- Blog list page: < 2 seconds
- Blog create form: < 1 second
- Blog save: < 3 seconds
- Blog detail page: < 2 seconds
- Image upload: 2-5 seconds (depending on size)
- Video upload: 10-30 seconds (depending on size)

Expected File Sizes:

- Optimized images: < 500KB (Cloudinary auto-optimizes)
 - Blog page HTML: < 100KB
 - CSS bundle: < 200KB
 - JS bundle: < 1MB
-

Next Steps After Testing

Once all tests pass:

1. Deploy to Staging:

- Push changes to staging environment
- Run full test suite again
- Test with real content

2. User Acceptance Testing:

- Have content team test blog creation
- Gather feedback on editor usability
- Make any UX improvements

3. Production Deployment:

- Deploy backend changes
- Deploy frontend changes
- Monitor for errors
- Watch Cloudinary usage

4. Documentation:

- Share TIPTAP_USAGE_GUIDE.md with content team
- Create video tutorial (optional)
- Update internal documentation

5. Monitoring:

- Check error logs
 - Monitor Cloudinary bandwidth
 - Track blog engagement metrics
-

Support

If you encounter issues not covered in this guide:

1. Check browser console for JavaScript errors
 2. Check backend logs for API errors
 3. Verify environment variables in `.env`
 4. Review `BLOG_IMPLEMENTATION.md` for technical details
 5. Check `TIPTAP_USAGE_GUIDE.md` for usage instructions
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Summary

The blog system now has full rich text editing capabilities with TipTap. All images and videos are automatically uploaded to Cloudinary and stored in organized folders. The frontend properly displays all rich content with beautiful, consistent styling.

Key Files Modified:

- `frontend/src/pages/static/BlogDetailPage.tsx` - Fixed HTML rendering
- `frontend/src/styles/index.css` - Added blog content styles
- `frontend/src/components/admin/BlogForm.tsx` - Already using TipTapEditor (from previous update)

Ready to Test! 